

The challenge we took on

Where should America's next sustainable data center be built - and how will that place feel about it?

THE CHALLENGE

The US is racing to site data centers for AI and cloud. Most tools rank cheap power and tax breaks. That is how we got Loudoun County, Virginia: a multi-year wait for the grid, and more than \$100 billion of projects blocked or delayed by local opposition since 2023.

A site that looks perfect on paper is worthless if it cannot get power or a permit. Sustainability is not one number. We score both "can it get built?" and "should it be built?"

TEAM SKYWALKERS

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GitHub org

github.com/AI-Data-Center-ML-Engine-SkyWalkers

Repos: siting-platform (app) and ML-Model (scoring)

WHAT WE BUILT

A working siting platform with three workspaces on one ranking: **Site finder** (globe + satellite map), **Trade-offs** (iMasons Social Accord), and **Community pulse** (live bills, news, and posts). FastAPI calls a seven-pillar scoring engine, then community signals re-rank the list.

1. THE HACKER MINDSET

Who we are and the bet we made

PEOPLE

Riddhi Raina Prasad -
rrp4822@nyu.edu - product,
interface, architecture

Krish Jani - kj2743@nyu.edu -
scoring model, research, feature
store

Passions: sustainable digital
infrastructure, honest
community signals, iMasons
Social Accord

Goals: ship a working app, not
a slide deck; keep every claim
tied to a public source

OPPORTUNITY

Vision: America sites the next
wave of data centers where the
grid, the water, and the people
can take them.

First idea: a map that ranks
counties on clean power.

What we added: permission
risk and live public sentiment -
the two things siting tools skip.

Why now: opposition groups
rose from 142 to 843 in about a
year. Projects die at the queue
and the council vote.

TEST

Assumption: projects fail on
climate or cost. **False.** 60 US
projects (2022-2026) fail at the
grid queue and the council vote.

Assumption: cheap-power
maps are enough. **False.**
Loudoun still "wins" those maps
and still cannot interconnect in
time.

Assumption: communities leave
a public trail. **True.** Bills, news,
and posts geotag to a county
FIPS.

Assumption: a black-box rank
will be trusted. **False** for this
buyer. They need a score they
can explain.

HACKATHON GOAL AND JUDGING

Ship rank, trade off, and watch a place - plus a scoring contract the ML repo can fill. We designed for technology, presentation, innovation, execution, and theme. Communication: GitHub org and the live app.

1. THE HACKER MINDSET

How we go, what we built, why it matters

GO

Customer: site engineer or real estate developer choosing a US data-center location.

Content: Rank every county on sustainability, then see how the community is moving before you buy the land.

Channels: globe and satellite map, shortlist, GitHub.

Call to action: open Site finder, move the sliders, click a point, read the pulse.

STORY

In our research we read blocked, withdrawn, and delayed US projects. They rarely die because the climate model was wrong. They die because the transmission never comes, or the county says no.

So we built a rank that treats permission and sentiment as first-class signals, not a footnote.

BUILD

- Site finder: 3D globe, Mapbox satellite, pillar sliders, click-to-zoom deep dive
- Trade-offs: iMasons Social Accord, 18 indicators, do-no-harm floor, radar
- Community pulse: news, bills, posts, Gemini summaries, US map, alerts
- FastAPI: scoring, re-rank, awareness, Social Accord engine
- ML path: 3,109 counties, about 100 features, seven-pillar MCDA, POST /rank

GROWTH / SPOT / OPTIMIZE

Horizon 1: live ML service on POST /rank, keep collecting.

Horizon 2: parcel and H3 cells; utility-commission dockets.

Horizon 3: a watched shortlist for a real developer portfolio.

Optimize: community re-rank after the model. A moratorium zeros a site. Robustness: 2,000 random weight sets. If a county stays top 10, it is stable.

2. RAPID IDEATION

What the customer feels, and the idea we kept

EMPATHY

I am a site engineer or real estate developer. **I am trying to** pick a US county for a large data center before we spend years on land and interconnection. **But** the maps I get are cheap-power heatmaps. **Because** the industry still treats community opposition and water as afterthoughts. **Which makes me feel** exposed: I can "win" a model and still lose the county hearing. **In a perfect world** I would see one shortlist that is honest about power, water, climate, land, and whether people will let this get built.

SEE

Queues, brownfields, protest headlines, state moratoria

THINK

Is this county actually buildable, or only cheap?

FEEL

Urgency from AI load, risk from a seven-year grid wait

DO

Compare a few states, fly a consultant, then get surprised by a bill

PAINS

Time-to-power, water stress, farmland fights, surprise moratoria, tools that hide a wrecked pillar behind a high average

GAINS

A shortlist that survives weight changes, a radar for economic / social / ecological trade-offs, a pulse that links every claim to a source

ROUND 1

Energy and carbon only: eGRID, time-to-power, surplus renewables

ROUND 2

Add water, hazard, land reuse, heat recovery. Still a static map.

ROUND 3

Permission risk from 60 failed projects and 31 moratoria, plus a live community engine. Rank, then re-rank.

FINAL IDEA AND MINIMUM AWESOME PRODUCT

SitewellEco2: score every US county with a transparent seven-pillar model, let public sentiment move the rank, and let the user trade off economic, social, and ecological value with a do-no-harm floor. Three workspaces on one ranking. Globe or satellite. Sliders. Accord radar. Live pulse.

3. HACKER'S CANVAS

The whole plan on one canvas

PROBLEM

Siting tools optimize for cheap power and tax breaks. They miss the two ways projects actually die: the grid queue and the council vote. A high score that hides terrible water or an active moratorium is dangerous.

EXISTING ALTERNATIVES

Utility heatmaps, consultant slide decks, county GIS, news search, and black-box models trained on where data centers already are (which just rediscovers Dallas, Houston, and Northern Virginia).

CUSTOMER SEGMENTS

Site engineers, real estate developers, and digital-infrastructure teams. Later: utilities and counties that want inbound interest shown honestly.

EARLY ADOPTERS

Developers comparing a shortlist of states right now. Teams already using iMasons language: People, Planet, Power, Pushback, Policy.

UNIQUE VALUE PROPOSITION

We do not claim the "right" weights. We show a top 10 that stays on top when weights move, then we let the user trade off live, and we re-rank when the community speaks.

HIGH-LEVEL CONCEPT

Like a siting heatmap - but it scores sustainability in the open, and it keeps listening after the shortlist is drawn.

KEY BENEFITS

3,109 counties, seven pillars, hard exclusions, geometric mean so one weak pillar cannot hide, community re-rank, iMasons trade-offs, every pulse item linked to its source.

TEAM TALENTS

Riddhi: product, React, Mapbox, globe, FastAPI wiring, architecture. Krish: geospatial features, MCDA scoring, research CSVs, robustness. Shared: Gemini awareness path, Social Accord.

CHANNELS

The live app. GitHub: [AI-Data-Center-ML-Engine-SkyWalkers](#). Hackathon demo. iMasons-aligned materials.

METRICS

Counties scored. Percent of top 10 that survive 2,000 random weight sets. Rank change after community re-rank. Coverage and confidence per county. Alerts from official or news sources only.

REVENUE ROADMAP

Hackathon: prove the method. Next: hosted shortlists for a developer team. Later: watched portfolios and county-level briefings.

COST TO DEVELOP

Open government data, research time on 60 projects, joins and robustness runs, Gemini summaries, a Mapbox token, engineering time for the three workspaces.

4. BUSINESS MODEL CPR

Who pays, and why we win

INDUSTRY

Digital infrastructure and data-center real estate

CUSTOMER

The person who has to recommend a county and defend it: site engineer or real estate developer

LOCATION

United States. Feature store: 3,109 counties. App explores candidates across the country.

POSITION

The sustainability and permission layer on top of power-price maps - not another "where the last campus went" model

OPPORTUNITY

Tens of billions in delayed or blocked load, and a Social Accord language the industry already accepted

X FACTOR

Community re-rank after a transparent score. A moratorium is a zero, not a footnote.

NETWORK

iMasons Social Accord. Public APIs (Open States, LegiScan, Federal Register, GDELT, Bluesky). One HTTP contract between app and ML: POST /rank.

MODEL

Near term: project-based scoring for a shortlist. Later: subscription to a watched set of counties.

MEASUREMENT

Robustness percent in top 10. Backcast of known failures. Community coverage. Time from slider move to new rank.

GO TO MARKET

Demo the three workspaces. Hand a developer their top 10 under two personas (carbon-first vs cost-first). Show a county that falls out when a moratorium appears.

EDUCATION

Architecture diagram and this toolkit. Every pulse card quotes the sentence the model used.

YOUR STRENGTHS

A working app, not a mockup. Open MCDA. Live awareness path. Research that explains the extra pillar: 60 failed projects, 31 moratoria.

5. PITCH LIKE A PRO

Why a site engineer would use this now

DESIRABILITY

Persona: site engineer / real estate developer

Problem: they need a place that can get power and a permit, not only a cheap megawatt.

Purpose: help them pick a county they can stand behind in a board room and a town hall.

Benefits: one shortlist, live weights, community movement, Accord trade-offs, sources they can click.

FEASIBILITY

Core team: Riddhi Raina Prasad and Krish Jani, SkyWalkers

Landscape: consultant GIS, utility maps, generic site-selection ML. None combine open county features, a veto, and a live social engine on iMasons terms.

We already run: FastAPI, React, Mapbox, globe, Gemini, SQLite, Parquet, NumPy.

VIABILITY

Revenue later: hosted shortlists and watched portfolios. Not the hackathon goal.

Roadmap: live ML service on POST /rank; more official dockets; parcel scale.

Re-iteration: bump model_version on every retrain. Re-rank betas live in env. Weights live in scoring.yaml.

THE PITCH

Most siting tools ask "where is power cheap?" That is how we over-built Northern Virginia and still cannot interconnect. We asked a harder pair of questions: **can it get built, and should it be built?**

We scored every US county on seven pillars - power, carbon, water, permission, hazard, land, and co-benefits - with hard exclusions and a geometric mean so cheap power cannot hide a dead water cycle. Then we keep watching. Legislation, news, and public posts go through Gemini. Features and alerts move the rank. Trade-offs uses the iMasons Social Accord so a gain in tax revenue cannot erase ecological harm.

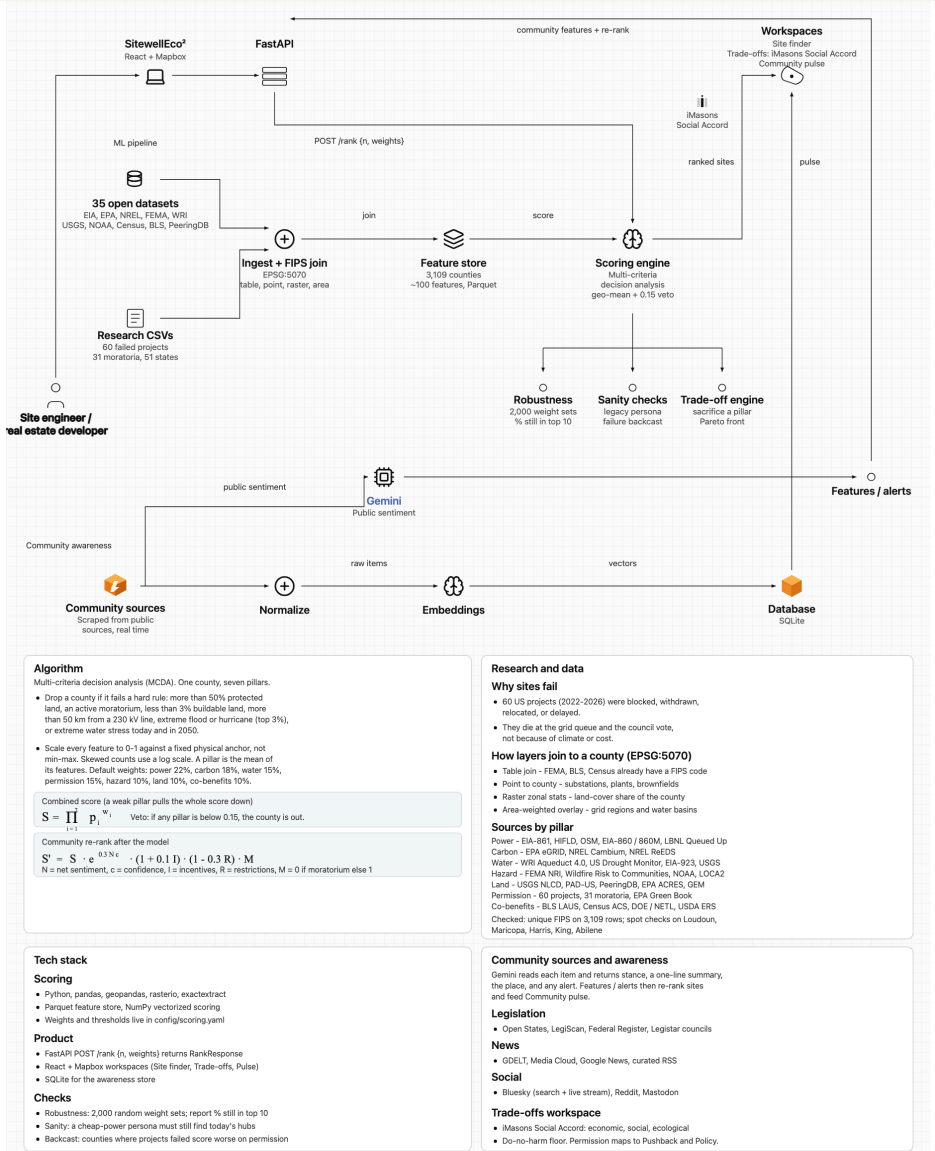
Open the globe. Move the sliders. Click a point. Read what that county is saying this week.

github.com/AI-Data-Center-ML-Engine-SkyWalkers

6. ARCHITECTURE

Two paths, one rank

A site engineer uses SitewellEco2 (React + Mapbox). FastAPI calls the scoring engine with POST /rank and the slider weights. Ranked sites open three workspaces. A second path scrapes public sources, stores them, and lets Gemini read public sentiment. Features and alerts flow back into FastAPI as a community re-rank, and into Community pulse.



The engine behind the map

ALGORITHM - MULTI-CRITERIA DECISION ANALYSIS

Not a black box. Rules a site engineer can explain.

- Drop a county if it fails a hard rule: more than 50% protected land, an active moratorium, less than 3% buildable land, more than 50 km from a 230 kV line, extreme flood or hurricane (top 3%), or extreme water stress today and in 2050.
- Scale every feature to 0-1 against a fixed physical anchor, not min-max. A pillar is the mean of its features.
- Default weights: power 22%, carbon 18%, water 15%, permission 15%, hazard 10%, land 10%, co-benefits 10%.

$$S = \text{product of } p_i^{w_i} \text{ from } i = 1 \text{ to } 7$$

A weak pillar pulls the whole score down. If any pillar is below 0.15, the county is out.

$$S' = S \cdot e^{0.3 N c} \cdot (1 + 0.1 I) \cdot (1 - 0.3 R) \cdot M$$

N = net sentiment, c = confidence, I = incentives, R = restrictions. M = 0 if moratorium, else 1.

TECH STACK

Scoring: Python, pandas, geopandas, rasterio, exactextract, Parquet, NumPy. Weights in config/scoring.yaml.

Product: FastAPI POST /rank returns RankResponse. React + Mapbox + globe. SQLite for awareness.

Checks: 2,000 random weight sets; a cheap-power persona must still find today's hubs; failed-project counties score worse on permission.

RESEARCH AND DATA

Why sites fail: 60 US projects (2022-2026) blocked, withdrawn, relocated, or delayed. They die at the grid queue and the council vote. That is why we added permission risk (iMasons Pushback and Policy).

Joins (EPSG:5070): table join (FEMA, BLS, Census), point to county (substations, plants, brownfields), raster zonal stats (land cover), area-weighted overlay (grid, basins).

About 35 open datasets become 3,109 counties x about 100 features (Parquet): EIA, EPA, NREL, FEMA, WRI, USGS, NOAA, Census, BLS, PeeringDB, plus research CSVs (60 projects, 31 moratoria, 51 states). Checked unique FIPS; spot checks on Loudoun, Maricopa, Harris, King, Abilene.

COMMUNITY SOURCES AND AWARENESS

Scrape first (no AI). Keyword gate, dedupe, geotag, store. Gemini writes stance, a one-line summary, place, and alerts. Embeddings power search. Features re-rank sites and feed pulse.

Legislation: Open States, LegiScan, Federal Register, Legistar

News: GDELT, Media Cloud, Google News, curated RSS

Social: Bluesky (search + live stream), Reddit, Mastodon

Official sources weigh more than social posts. A single social post never raises an event alert. A moratorium needs an official or established news source. Every item links to its source.

8. SCALE

How SitewellEco2 stays cheap as it grows

SCORE ONCE, RE-RANK OFTEN

The expensive work is offline: join 35 datasets to FIPS, write a 3,109 by 100 Parquet table. A slider move is a NumPy pass over that table, not a GIS rebuild. FastAPI is stateless. POST /rank is the contract, so a mock today and a live ML service tomorrow do not change the app.

ONE RANKING, THREE ROOMS

Site finder, Trade-offs, and Community pulse read the same shortlist. A new workspace does not multiply compute. Weights live in scoring.yaml. model_version bumps on retrain. Re-rank betas live in env, so we can tune sentiment without redeploying the feature store.

AWARENESS IS INCREMENTAL

Collectors paginate public APIs. We store first, then run Gemini only on new items. Keyword gate and dedupe keep the queue small. Embeddings are for search, not for every rank call. A scheduler can pull hourly without touching the county matrix.

SAME PATTERN, FINER GRAIN

County FIPS is today's grain. The same ingest and join pattern (EPSG:5070, table / point / raster / overlay) can drop to parcels or H3 later. New sources plug in behind normalize, geotag, store. SQLite is the hackathon store; the schema can move to Postgres without changing the product contract.

MORE USERS, SAME MATRIX

A watched portfolio is a filter on the same 3,109 rows plus a county alert feed. Robustness (2,000 weight sets) is an offline check, not a request-time cost. Official sources weigh more than social, so volume on Bluesky does not blow up the rank.

WHAT WE DO NOT PRETEND

We do not scrape the web on every click. We do not re-join rasters when a user moves a slider. We do not let one social post raise an event alert. Scale comes from a fixed feature store, a thin HTTP contract, and a pipeline that only spends AI on what is new.

TEAM AND LINKS

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<https://github.com/AI-Data-Center-ML-Engine-SkyWalkers>